

The 24th CIRP Conference on Life Cycle Engineering

Product Redesign for Improved Value Recovery via Disassembly Bottleneck Identification and Removal

Liang Cong^{a*}, Fu Zhao^{a,b}, John W. Sutherland^b^a Mechanical Engineering, Purdue University, West Lafayette, IN 47907, USA^b Environmental and Ecological Engineering, Purdue University, West Lafayette, IN 47907, USA* Corresponding author. Tel.: 1-765-430-8857; fax: 1-765-494-4482. E-mail address: lccong@purdue.edu

Abstract

Making progress towards a circular economy requires the reuse of material/components to realize their maximum utility. Integrating end-of-life (EOL) considerations into the design of products will greatly facilitate the recovery of material/components at EOL, and enable their reintroduction into new product life cycles. Past research has largely employed qualitative or single objective oriented approaches for environmental product design. This paper presents a quantitative method for value recovery, in which disassembly bottlenecks are identified and removed by improving upon an existing design. The approach determines the value that can be recovered from EOL products through different EOL options. For a given EOL scenario, major bottlenecks are generally associated with joints, material incompatibility, and product architecture. Once the bottlenecks to rapid, cost effective dismantling of EOL products are identified, different design changes are considered to reduce the time/cost required to recover components/materials. To demonstrate the effectiveness of the method, a hard disk drive is used as a case study.

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Peer-review under responsibility of the scientific committee of the 24th CIRP Conference on Life Cycle Engineering

Keywords: Design for Environment; Circular Economy; Recycling

1. Introduction

The circular economy principle requires the reuse of product end-of-life (EOL) materials/components to obtain maximum utility [1]. At present, product designers rarely give much consideration to how products will be managed at their EOL. Integrating EOL considerations into product design will greatly facilitate the value recovery of products at the end of their working lifespan.

There is a history of design philosophies, principles, and practices that seek to address environmental considerations during product design, e.g., “design for environment,” “eco-design,” “design for recycling,” and “design for disassembly” [2-10]. Often, these methods focus on the use stage of the product, and give little attention to the product EOL. When the product EOL stage is considered, emphasis is frequently limited to recycling, as opposed to recovering the maximum value from the used product. Moreover, many of these design methodologies take the form of guidelines and semi-quantitative methods.

Guidelines are generally qualitative principles and are widely utilized in industry. In the present context, guidelines are often suggestions that can be used to promote the recyclability of the components within a product. For instance, Dowie and Simon developed guidelines that covered three categories: materials, fasteners & connections, and product structure [11]. Rose et al. proposed a design advisor that guides designers to specify the end-of-life strategies [12]. Thorn developed a tool to assist designers to select product retirement strategy based on environment implication [13]. Sample guidelines include “Use materials which can be recycled,” and “Fasteners should be easy to remove.” A challenge associated with using guidelines to promote value recovery is the lack of clear operational definition. Words like “easy” and “good” have no communicable meaning, i.e., they lack the specificity needed for industrial implementation.

A common feature of semi-quantitative methods is that they seek to transform experiential or subjective observations

into engineering characteristics that are represented quantitatively. Quality function deployment (QFD) is an example of a semi-quantitative method that may be used to support eco-design. Devanathan et al. developed a semi-quantitative eco-design methodology that can be used in the early design stages, using a combination of QFD and life cycle analysis [14]. QFD requires that designers determine which parts are most environmentally important and improve associated design features [15]. One drawback is that correlations established between environmental performance and engineering characteristics are based on experience or engineering judgment as opposed to rigorous analysis. Another drawback is the lack of solid links between design changes and environmental performance.

When EOL value recovery is considered, material, joint, and part configuration are often the targeted design features to be improved [16-19]. In the EOL stage, products will be dismantled into several modules in a certain sequence to obtain maximum profit. Recycling rate of products will be increased, if eliminating the incompatibility of material composition of each module that will be recycled for material can be done in design stage. Choosing easily detachable joints reduces dismantling time. Last but not least, ease of access to reusable components will minimize the chance that the components will be damaged during disassembly.

Unlike the impact of guidelines and semi-quantitative methods, the impact of designing features on EOL value recovery can be quantified. By quantifying, the ease with which a given design may be recovered, reused, remanufactured, or recycled, the environmental performance of a product may be evaluated, and the best candidate among alternative designs may be selected. For instance, a model was developed to calculate the recyclable mass of consumer electronic products and the economic value of any recovered materials [20]. For another instance, a product configuration optimization method was developed for application to EOL disassembly [21]. These methods are single design feature oriented, and other important design features are neglected.

As noted, it is necessary to improve multiple design features and quantify the effect of the change on value recovery. To fill the gap, this paper proposes a quantitative approach that considers design features: joints, materials, and product configuration. The approach begins by modeling the EOL product value recovery process using mathematical programming, then continues with removing disassembly bottlenecks based on EOL scenario analysis, and evaluates the value recoverability of a new design. A case study for a hard disk drive (HDD) is conducted to demonstrate the approach.

2. A Model for EOL Value Recovery

As stated in the authors' previous work, knowledge of key product, process and market information is essential to construct a model for EOL value recovery [22, 23]. Product information provides spatial constraints among components and material composition. The value of recycled or disposed components is reflected through the market demand and price. Process information shows existing technology and

what tools are available to disassemble a product of interest. An example product is shown in Figure 1. The product consists of three parts A, B, C with joints 1 and 2. To proceed, information should be collected about material composition of each part, and what tools can be used to easily detach joints 1 and 2. In addition, check out possible EOL options of part A, part B, part C, component AB, and Component BC. The reason of investigation of components AB and BC is that each component may be the final EOL module for recycling or reuse. The definition of transition matrix describes all the possible dismantling operations when an EOL product is disassembled. To illustrate how to generate the transition matrix, take the example of removing A away from ABC (operation d1). In the column of d1, -1 indicates that ABC is disassembled and 1 means that BC and A are generated. The rest of entries are filled with zeros. "d0" is always to be conducted, which means that the entire product is given and will be shredded or disposed as a whole or disassembled to smaller modules. The definition of succession matrix describes flows of operations. In the case of the example product, the flows might be from d1 to d2 or from d2 to d1. The cost of dismantling is composed of base operation and transition cost coming from detaching joints, and changing tools and direction during the flow of operations.

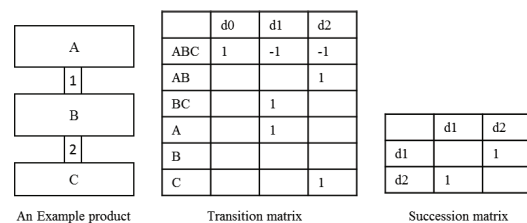


Fig. 1 An example product, transition matrix and succession matrix

A mathematical model for the dismantling may be constructed. The formation of problem is as follows.

Variables and Parameters:

- 1) I is the set of feasible subassemblies. The index i is used to refer to a specific subassembly.
- 2) J is the set of feasible disassembly operations. The index j is used to refer to a specific operation.
- 3) C is the cost matrix whose elements C_{jk} representing the transition cost from operation j to operation k .
- 4) E is the end-of-life option revenue matrix with elements E_{il} representing the revenue when subassembly i is processed to produce end-of-life option l .
- 5) S is the succession matrix with elements S_{jk} . $S_{jk} = 1$ means that operation k can follow operation j . Otherwise, operation k cannot be conducted after operation j .
- 6) T is the transition matrix with elements T_{ij} . The possible values for T_{ij} are -1, 1, and 0. When $T_{ij} = -1$, the j th operation will disassemble the i th subassembly. When $T_{ij} = 1$, it indicates that the i th subassembly will be released by the j th disassembly operation. The initial operation is to assume that a

product is given to recycle. Thus $T_{10}=1$, $T_{i0}=0$, where row 1 denotes the entire product.

- 7) M is the subset of the operations of transition matrix T that includes every operation except the initial one. The index m is used to refer to the basic operation of detaching joints.
- 8) C_B is the cost vector with elements C_{Bj} representing the cost of operation j .
- 9) C_T is the cost vector with elements C_{Tj} representing the cost of used tool in the operation j .
- 10) L is the lifespan vector with elements L_j representing normal working hours (lifespan) of tool associated with operation j .
- 11) F is the time vector with elements F_j representing the base time of operation j .

Decision variables:

- 1) Aggregate flow variables x_j , which are 0 or 1 variables.
- 2) Partial flow variables w_{jk} , which are 0 or 1 variables.
- 3) EOL option of each subassembly y_{il} which are 0 or 1 variables.

As an objective, it is desired to maximize the profit function below:

$$\sum_i \sum_j (\sum_l T_{il} E_{il} y_{il} x_j - \sum_k C_{jk} w_{jk} - C_{Bj} x_j - C_{Tj} F_j / L_j x_j)$$

The first term denotes the revenue from subassembly i disassembled via operation j and processed to form EOL option l . The second term is the cost from operation transition from j to k . The third term determines cost from base operation j . The last term represents facility cost.

$$x_k = \sum_j w_{j,k} \quad (1)$$

This equation establishes the relation between x_j and w_{jk} , which guarantees that the aggregate flow variable is set to 1 only if one of the relevant partial flow variable equals 1.

$$S_{j,k} - w_{j,k} \geq 0 \quad (2)$$

The equation eliminates infeasible transitions between operations.

$$\sum_j T_{ij} x_j \geq 0 \quad (3)$$

This equation makes sure that each subassembly is disassembled only after the subassembly is generated.

$$\sum_k w_{k,m} \geq \sum_k w_{m,k} \quad (4)$$

This equation guarantees that inflow and outflow are balanced for each operation.

$$\sum_m w_{j,m} \leq 1 \quad (5)$$

This equation makes sure that outflow from each operation is allowed less than or equal to once.

$$\sum_l y_{il} \leq 1 \quad (6)$$

This equation guarantees that each subassembly cannot be reprocessed in more than one EOL option.

The linear model will be solved to find out EOL value recovery plan: the economic profit, the optimal disassembly sequence, disassembly level (EOL modules) and resultant EOL option of each EOL module.

3. Bottleneck Identification and Redesign.

As has been noted, economic profit is the principal motivation for recovering and recycling EOL products. Without integrating EOL considerations into product design, the recycling cost will be much larger than it could be, and may require time-consuming breaking of joints, material incompatibility that leads to excessive sorting in preparation for recycling, and difficulty in extracting valuable components due to product architecture/configuration constraints. When the dismantling cost is inflated, it may dominate the revenue – and owing to challenges in recovery, revenue opportunities may be compromised as well. Lack of EOL thinking results not only in low economic returns, but also may provide little recovered value in terms of component function and material. The aim of the paper is to remove the bottleneck to recycling and change associated design features.

Associated design features include joints, materials, part configuration. Obviously, easily released joints allow other components to be recovered without profit and time compromise. Material compatibility will ensure the profitability of recycling EOL modules, instead of disposing them. Reusable and hazardous components should be easily accessed to ensure the quality of reusable components and safety of operation.

Analysis of the EOL scenario and identification of bottlenecks are important to make changes in design features for improved value recovery. For identification of major bottleneck coming from detaching joints, Pareto principle (also called 80-20 rule) is applied. A Pareto analysis can be performed by constructing a histogram of the costs associated with various disassembly tasks. These costs would be ordered from highest to lowest cost. The same chart is added by a cumulative percentage curve and a horizontal 80% line. A vertical line can be added, passing through the intersection of the cumulative percentage curve and the 80% line. The major causes of cost are from the operations on the left side of the vertical line. Often, the majority of the costs are associated with one or two disassembly tasks. As mentioned early, the cost of dismantling is composed of base operation and transition cost coming from detaching joints, and changing tools and direction during the flow of operation. So the Pareto analysis may be applied to base cost and transition cost. The results of the analysis will reveal the targeted joints need to be replaced. The criterion of joint replacement is detaching time. Among alternative joints, the best candidate is the one with least detaching time. For identification of major bottlenecks to material recycling, EOL modules composed of more than two parts will be checked for material compatibility. Plastic and metallic material compatibilities are shown in references [24] and [25] respectively. Incompatible materials will be changed, according to material compatibility tables in the references. The priority of component reconfiguration is based on the ratio of EOL revenue to disassembly cost. Each EOL module will be evaluated according to the ratio. The EOL module with the top ratio value will be reconfigured according to opportunities.

4. Case study

A personal computer hard disk drive (HDD) is used to demonstrate the effectiveness of the proposed approach. Figure 2 shows a disassembled HDD with its components identified and labelled. The HDD architecture and connections between components are indicated in Figure 3.

The transition matrix for the indicated HDD was constructed and is shown in Table 1. In the table, all the entries not displayed have values of zero. Possible EOL options for the subassemblies/components and their associated revenues are shown in Table 2. Estimation of revenue is based on market information. An entry of $-\infty$ indicates that the subassembly/component will not be treated in any form of EOL option. The tools used for each operation/task, task time, and cost are shown in Table 3. T1-T5 represent different sized screwdrivers, T0 indicates hand, and TN denotes a combination of tools used to release the component/subassembly. As mentioned, during the discussion of the mathematical model, the succession matrix indicates the feasibility of flow between operations. The succession information is shown in Table 4. In Table 4, unfilled entries have a value of zero. Filled entries are in the format of 1/xx. An entry of “1” means that the flow from the row operation to the column operation is feasible. XX is the cost (in dollars) associated with the flow.

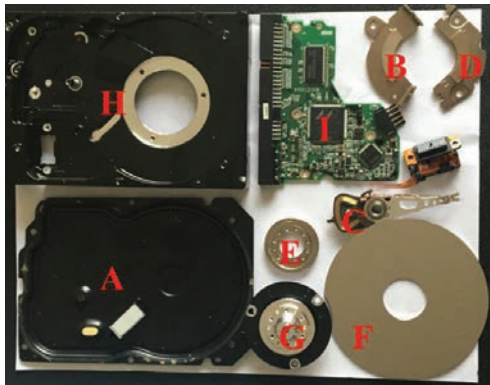


Fig. 2 A disassembled HDD

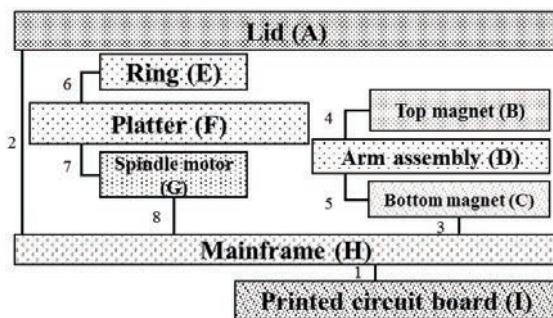


Fig. 3 Diagram of HDD architecture

Table 1. Transition matrix

	d0	d1	d2	d3	d4	d5	d6	d7	d8
A-I	1	-1							
A-H		1	-1						
B-H			1	-1					
B-D				1	-1				
E-H				1			-1		
F-H							1	-1	
GH								1	-1
CD					1	-1			
A			1						
B				1					
C					1				
D						1			
E							1		
F								1	
G									1
H									1
I		1							

Table 2. Subassembly EOL option and revenue

	Reuse (\$)	Recycling (\$)	Disposal (\$)
A-I	$-\infty$	$-\infty$	-0.005
A-H	$-\infty$	$-\infty$	-0.004
B-H	$-\infty$	$-\infty$	-0.004
B-D	$-\infty$	$-\infty$	-0.001
E-H	$-\infty$	$-\infty$	-0.003
F-H	$-\infty$	$-\infty$	-0.003
GH	$-\infty$	$-\infty$	-0.003
CD	$-\infty$	$-\infty$	-0.001
A	$-\infty$	0.021	-0.001
B	0.800	0.004	-0.001
C	0.800	0.004	-0.001
D	$-\infty$	0.006	-0.001
E	$-\infty$	0.001	-0.001
F	$-\infty$	$-\infty$	-0.001
G	2.000	0.012	-0.002
H	$-\infty$	0.050	$-\infty$
I	$-\infty$	0.355	-0.001

Table 3. Operation tool, time and tool cost

	d1	d2	d3	d4	d5	d6	d7	d8
Time(s) / Labor Cost (\$)	40s/ 0.13	70s/ 0.23	50s/ 0.17	15s/ 0.05	25s/ 0.08	60s/ 0.2	5s/ 0.02	30s/ 0.1
Tool Code/ Cost (\$)	T4/ 2	T1/2	TN/2	T1/2	T3/2	T2/ 2	T0/ 0	T5/ 2
Tool lifespan (h)	200	200	200	200	200	200	$+\infty$	200

Table 4. Succession information

	d1	d2	d3	d4	d5	d6	d7	d8
d1		1/0.02						
d2			1/0.01					
d3				1/0.01		1/0.01		
d4					1/0.01	1/0.02	1/0.02	1/0.02
d5						1/0.02	1/0.02	1/0.02
d6				1/0.02	1/0.02		1/0.01	
d7				1/0.02	1/0.02			1/0.01
d8				1/0.02	1/0.02			

The final maximized profit of recycling the HDD is \$2.96. Disassembly operations are in the order of 0, 1, 2, 3, 4, 5, 6, 7, 8, which means that maximum profit is obtained after complete disassembly. The spindle motor will be reused and other parts will be recycled for the material except for the platter, which will be discarded. The change of revenue and operation cost with proceeding of operations is shown in Figure 4.

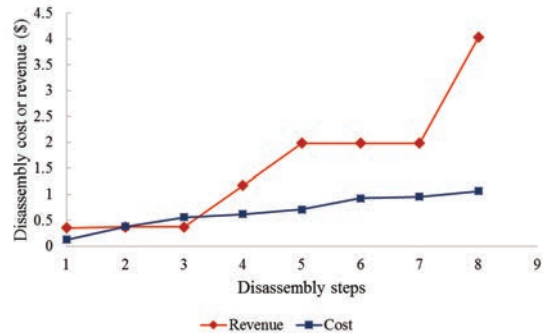


Fig. 4 Change of revenue and cost with proceeding of operations

To identify the major bottlenecks to dismantling, the operation cost is categorized into a base operation cost and a transition cost. Transition cost comes from transition or flow of operations, resulting from the changes of tool and operation direction. The analysis of the EOL scenario shows that transition cost takes up about 8% cost in this case. 92% cost results from base disassembly operations. In comparison, base cost is much higher than transition cost. Also, it means that detaching joints is cost intensive.

The Pareto principle is used to identify which base operations constitute the majority of the cost. A Pareto chart for the base cost is shown in Figure 5. As is evident, the overwhelming majority of the cost is associated with operations 2, 6, 3, and 1. Operation 2 is releasing the lid from the mainframe, operation 6 is releasing the ring from the platter, operation 3 is separating two subassemblies, and operation 1 is releasing the printed circuit board. Reducing the number of screws and changing the type of joints have great potential to maximize profit. If this approach is applied in other cases, it can be predicted that more parts can be recovered at EOL. In this case, maximum profit is obtained when products are disassembled to individual components. So no more parts need to be released and recovered.

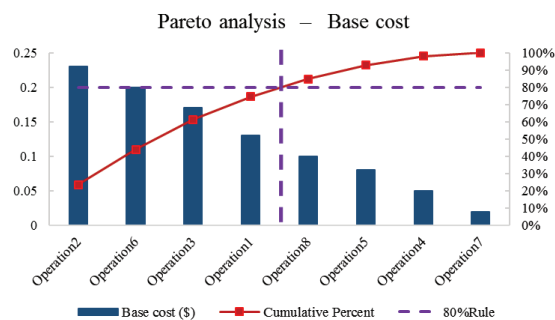


Fig. 5 Pareto analysis on base cost

Transition cost is mostly due to 5 different sizes of screws. Reducing the number of screws sizes will help minimize cost and reduce operation effort needed to maximize the profit.

In this case study, the product is disassembled to individual components. Most parts are recyclable. No EOL module consists of more than one part. If EOL module is made of more than one part, the material compatibility should be examined for the module in order to increase recovered profit or material value.

For this HDD, the spindle motor has the highest ratio of EOL revenue to disassembly cost and determined to be re-positioned. For the sake of ease of removal and to ensure its quality, it might be suggested that the spindle motor be placed in the other side of the mainframe.

In response to analysis of this HDD EOL scenario, to maximize EOL value recovery, it is decided that snap-fit is best used to replace screws connecting the lid and mainframe. Using snap-fit, the disassembly time can be reduced from 70s to 4s. The number of screws then connecting the ring and mainframe is reduced to 3, with 50% disassembly time reduction. The same size of screws should be used throughout the design. The spindle motor will be placed in the opposite side of HDD. These design modifications will result in the change of transition matrix, succession matrix and other process information. After changes in product and process information were inputted, the mathematical model is executed again. The result shows improved profitability of recycling the new product. The profit of recycling a newly designed HDD will be \$3.27, as compared to \$2.96 from the original HDD.

5. Summary and conclusions

A quantitative redesign approach is proposed for improved value recovery. The approach is capable of measuring the recyclability and profitability of a current product design through modeling EOL value recovery. With analysis of an EOL recycling scenario, bottlenecks to value recovery can be identified for such aspects as joints, materials, and product configuration. Identifying and removing value recovery bottlenecks through an improved design can not only reduce dismantling costs (and increase profits), but also has a greater potential to recover more function and material value, which will then increase the revenue associated with recovered components (also increasing profit).

The HDD EOL scenario demonstrates that reducing the overall number and type of screws and locating the reusable valuable components in easily accessible positions can greatly increase profitability and ease of value recovery.

In relation to previous research, the primary merit of the approach is to allow designers to identify and remove the major bottlenecks in various key design features through modelling value recovery based on product architecture, dismantling and recycling technologies, secondary market demand. The result of the case study reveals that profitability of recovering the value of components can be improved through redesign. Therefore, instead of shredding HDDs, reusing components and material can become a primary

means for recyclers to support circular economy development.

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